

Modern methods in Harmonic Analysis

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2025 Fall

Preface

These notes are based on the short course given by Joshua Zahl during 2025 Fall at Nankai University. The content is mainly about Kakeya and restriction theory, with the emphasis of introducing modern methods in harmonic analysis. Typos or any nontrivial mistakes in these notes should be attributed to the author, since the lecturer has made his talks sufficiently clear.

Hope you find these notes enjoyable to read!

Nankai University, January 2026,
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1 Multilinear Keakeya

There are mainly two reasons of introducing multilinear Keakeya. One is that though it seems like a generalization of linear Keakeya judging from the name, it is actually easier to deal with and helps the proof of linear Keakeya. The second reason is that the technique of induction on scales used in the proof of multilinear Keakeya plays an important role in many other contexts in harmonic analysis. First we state the multilinear Keakeya theorem of Bennett- Carbery- Tao:

Theorem 1.1. (Bennett- Carbery- Tao) *Let $\varepsilon, \delta > 0$, $\mathbb{T}_1, \dots, \mathbb{T}_n$ be (finite) sets of δ -tubes in $\mathbb{R}^n$. Suppose that each tube $T_i \in \mathbb{T}_i$ points in the direction making angle $\leq \frac{1}{100n}$ with e_i , where e_i is the direction of the i -th coordinate axis in $\mathbb{R}^n$. Then*

$$\left\| \prod_{j=1}^n \left(\sum_{T \in \mathbb{T}_j} \chi_T \right) \right\|_{L^{1/(n-1)}(B(0,1))} \leq C_{n,\varepsilon} \delta^{-\varepsilon} \prod_{j=1}^n (\delta^{n-1} \# \mathbb{T}_j). \quad (1)$$

We will first prove Theorem 1.1 for the very special case where tubes in $\mathbb{T}_j$ point precisely in the direction e_j . Then we will reduce Theorem 1.1 to the case where tubes in $\mathbb{T}_j$ point almost (depending on ε) in the direction e_j , and apply induction on scales.

Lemma 1.1. (Loomis-Whitney) *Suppose $f_1, f_2, \dots, f_n \in L^{n-1}(\mathbb{R}^{n-1})$, $n \geq 2$. Then we have*

$$\int_{\mathbb{R}^n} \left| \prod_{i=1}^n f_i \circ \Pi_i \right| \leq \prod_{i=1}^n \|f_i\|_{L^{n-1}}, \quad (2)$$

where Π_i is the projection from $\mathbb{R}^n$ to $\mathbb{R}^{n-1}$ that forgets the i -th coordinate.

Proof. For $n = 2$, (2) is trivial by Tonelli. Suppose (2) holds for $k \leq n$. For the case $n + 1$,

$$\begin{aligned} \text{LHS of (2)} &= \int_{\mathbb{R}^n} f_{n+1}(x_1, \dots, x_n) \left(\int_{\mathbb{R}} \prod_{i=1}^n |f_i \circ \Pi_i| dx_{n+1} \right) dx_1 \cdots dx_n \\ &\leq \|f_{n+1}\|_{L^n} \left(\int_{\mathbb{R}^n} \left(\int_{\mathbb{R}} \prod_{i=1}^n |f_i \circ \Pi_i| dx_{n+1} \right)^{\frac{n}{n-1}} dx_1 \cdots dx_n \right)^{\frac{n-1}{n}} \\ &\leq \|f_{n+1}\|_{L^n} \left(\int_{\mathbb{R}^n} \prod_{i=1}^n \left(\int_{\mathbb{R}} |f_i \circ \Pi_i|^n dx_{n+1} \right)^{\frac{1}{n-1}} dx_1 \cdots dx_n \right)^{\frac{n-1}{n}}, \end{aligned}$$

by Holder's inequality. Let $g_i(x_1, \dots, \hat{x}_i, \dots, x_n) = \left(\int_{\mathbb{R}} |f_i \circ \Pi_i|^n dx_{n+1} \right)^{\frac{1}{n-1}}$ (note that this

is independent of x_i). By induction hypothesis, we have

$$\begin{aligned} & \|f_{n+1}\|_{L^n} \left(\int_{\mathbb{R}^n} \prod_{i=1}^n g_i(x_1, \dots, \hat{x}_i, \dots, x_n) dx_1 \cdots dx_n \right)^{\frac{n-1}{n}} \\ &= \|f_{n+1}\|_{L^n} \prod_{i=1}^n \|g_i\|_{L^{\frac{n}{n-1}}}^{\frac{n-1}{n}} \\ &= \prod_{i=1}^{n+1} \|f_i\|_{L^n} = \text{RHS of (2)}. \end{aligned}$$

This concludes the induction. $\square$

Proposition 1.1. Let $\varepsilon, \delta > 0$, $\mathbb{T}_1, \dots, \mathbb{T}_n$ be finite sets of δ -tubes in $\mathbb{R}^n$. Suppose that each tube $T_i \in \mathbb{T}_i$ points in the direction e_i . Then

$$\left\| \prod_{j=1}^n \left(\sum_{T \in \mathbb{T}_j} \chi_T \right) \right\|_{L^{1/(n-1)}(B(0,1))} \leq C_{n,\varepsilon} \delta^{-\varepsilon} \prod_{j=1}^n (\delta^{n-1} \# \mathbb{T}_j). \quad (3)$$

Proof. Let $f_j = \left(\sum_{T \in \mathbb{T}_j} \chi_{\Pi_j(T)} \right)^{\frac{1}{n-1}} \chi_{B_{n-1}(0,1)}$, $j = 1, \dots, n$, where $B_{n-1}(0,1)$ stands for the $(n-1)$ -dimensional unit ball. Apply Loomis-Whitney (Lemma 1.1). $\square$

Our first big task would be to deduce Theorem 1.1 from the following lemma:

Lemma 1.2. For any $\varepsilon > 0$, there exists $C'_{n,\varepsilon} > 0, \rho > 0$ such that if $\mathbb{T}_j$ is any set of tubes which make angle $\leq \rho$ with e_j , $j = 1, \dots, n$, then

$$\int_{Q_S} \prod_{j=1}^n \left(\sum_{a=1}^{N_j} \chi_{T_{j,a}} \right)^{1/(n-1)} \leq C'_{n,\varepsilon} S^\varepsilon \prod_{j=1}^n N_j^{1/(n-1)}, \quad (4)$$

where $N_j = \# \mathbb{T}_j$, $\mathbb{T}_j = \{T_{j,1}, T_{j,2}, \dots, T_{j,N_j}\}$, $Q_S \subseteq \mathbb{R}^n$ is any cube of side length S , $S \geq 1$.

Proof of Theorem 1.1. Suppose Lemma 1.2 is verified to be true. Given $\mathbb{T}_1, \dots, \mathbb{T}_n$ as in the hypothesis of Theorem 1.1. Let $\varepsilon > 0$, inducing $C'_{n,\varepsilon}, \rho$ from Lemma 1.2. For each j , the direction of tubes in $\mathbb{T}_j$ are contained in a spherical cap $S_j \subseteq S^{n-1}$ of radius $\frac{1}{100n}$. Cover S_j by $C_n \left(\frac{1/100n}{\rho} \right)^{n-1} \lesssim_{n,\varepsilon} 1$ many caps $\{S_{j,\beta}\}_\beta$ of radius ρ , and put $\mathbb{T}_{j,\beta} := \{T \in \mathbb{T}_j : \text{dir}(T) \in S_{j,\beta}\}$. Then, by Minkowski's inequality and Lemma 1.2, we

have

$$\begin{aligned}
\int_{Q_S} \prod_{j=1}^n \left(\sum_{T \in \mathbb{T}_j} \chi_T \right)^{\frac{1}{n-1}} &= \int_{Q_S} \left(\sum_j \sum_{S_{j,\beta}} \prod_{j=1}^n \left(\sum_{T \in \mathbb{T}_{j,\beta}} \chi_T \right) \right)^{\frac{1}{n-1}} \\
&\leq \sum_j \sum_{S_{j,\beta}} \int_{Q_S} \left(\prod_{j=1}^n \left(\sum_{T \in \mathbb{T}_{j,\beta}} \chi_T \right) \right)^{\frac{1}{n-1}} \\
&= \delta^n \sum_j \sum_{S_{j,\beta}} \int_{Q_S} \left(\prod_{j=1}^n \left(\sum_{T \in \mathbb{T}_{j,\beta}} \chi_{\delta^{-1}T} \right) \right)^{\frac{1}{n-1}} \\
&\lesssim_{n,\varepsilon} S^\varepsilon \prod_{j=1}^n (\delta N_j^{\frac{1}{n-1}}),
\end{aligned}$$

for any cube $Q_S \in \mathbb{R}^n$ of side length $S \geq 1$. Taking Q_S to be the unit cube containing $B(0, 1)$ completes the proof. $\square$

To prove lemma 1.2, we need the technique of induction (on the scale of tubes). Some notations are needed for the convenience of statements.

Given a set $\mathbb{T}_j$ of tubes, a positive number $W \geq 1$ (independent of δ), let $T_{j,a,W}$ be the W -neighborhood of $T_{j,a}$, $a = 1, \dots, N_j$. Put

$$f_{j,W} := \sum_{a=1}^{N_j} \chi_{T_{j,a,W}},$$

the sum of characteristic functions of W -thickened tubes in $\mathbb{T}_j$.

Lemma 1.3. *Suppose $\mathbb{T}_j$ is a set of tubes making angle $\leq \rho$ with the direction e_j , $j = 1, \dots, n$, $\rho > 0$. Let $W \geq 1$. If $S \geq \rho^{-1}W$, then*

$$\int_{Q_S} \prod_{j=1}^n f_{j,W}^{1/(n-1)} \leq C_n \rho^n \int_{Q_S} \prod_{j=1}^n f_{j,\rho^{-1}W}^{1/(n-1)}. \quad (5)$$

Proof. Divide Q_S into cubes Q of side length between $\frac{1}{20}\rho^{-1}W$ and $\frac{1}{10}\rho^{-1}W$. It suffices to prove (5) with Q in place of Q_S . The key observation is that, for any $T_{j,a,W}$, there is a tube $\tilde{T}_{j,a,W}$ of thickness $2W$, pointing in the direction e_j , such that $T_{j,a,W} \cap Q \subseteq \tilde{T}_{j,a,W} \cap Q$. Hence we have

$$\int_Q \prod_{j=1}^n f_{j,W}^{\frac{1}{n-1}} \leq \int_Q \prod_{j=1}^n \left(\sum_{\substack{1 \leq a \leq N_j \\ T_{j,a,W} \cap Q \neq \emptyset}} \chi_{\tilde{T}_{j,a,W}} \right)^{\frac{1}{n-1}}. \quad (6)$$

Now we can apply Loomis-Whitney (Proposition 1.1) on RHS of (6). Let $N_j(Q)$ denote the number of tubes in $\mathbb{T}_{j,W} := \{T_{j,a,W} : 1 \leq a \leq N_j\}$ that intersect Q . Then, by rescaling

by a factor of W^{-1} to tubes of thickness 1, RHS of (6) is

$$\leq C_n W^n \prod_{j=1}^n N_j(Q)^{\frac{1}{n-1}}.$$

To finish the proof, we need to compare $N_j(Q)$ with RHS of (5). Note that if $T_{j,a,W} \cap Q$ is nonempty, then $Q \subseteq T_{j,a,\rho^{-1}W}$. Therefore,

$$\prod_{j=1}^n N_j(Q)^{\frac{1}{n-1}} \leq \prod_{j=1}^n \left(\sum_{a=1}^{N_j} \chi_{T_{j,a,\rho^{-1}W}}(x) \right)^{\frac{1}{n-1}}$$

for all $x \in Q$. This implies

$$\begin{aligned} C_n W^n \prod_{j=1}^n N_j(Q)^{\frac{1}{n-1}} &\leq C_n W^n \frac{1}{|Q|} \int_Q \prod_{j=1}^n f_{j,\rho^{-1}W}^{\frac{1}{n-1}} \\ &\lesssim C_n \rho^n \int_Q \prod_{j=1}^n f_{j,\rho^{-1}W}^{\frac{1}{n-1}}, \end{aligned}$$

noting that $|Q| \sim (\rho^{-1}W)^n$. This completes the proof. $\square$

Now we can prove Lemma 1.2 using induction on scale (Lemma 1.3).

proof of Lemma 1.2. If $S = \rho^{-M}$ for some $M \in \mathbb{Z}_{>0}$, where ρ is an undetermined small positive number. Apply induction-on-scale lemma M times:

$$\begin{aligned} LHS \text{ of (4)} &\sim_n \int_{Q_S} \prod_{j=1}^n f_{j,1}^{\frac{1}{n-1}} \\ &\leq C_n \rho^n \int_{Q_S} \prod_{j=1}^n f_{j,\rho^{-1}}^{\frac{1}{n-1}} \\ &\leq \dots \leq C_n^M \rho^{Mn} \int_{Q_S} \prod_{j=1}^n f_{j,\rho^{-M}}^{\frac{1}{n-1}} \\ &\lesssim_n C_n^M \sup \left(\prod_{j=1}^n f_{j,\rho^{-M}}^{\frac{1}{n-1}} \right) \\ &\leq C_n^M \prod_{j=1}^n N_j^{\frac{1}{n-1}}. \end{aligned}$$

Pick $\rho \leq C_n^{-\frac{1}{\varepsilon}}$. The estimation above yields

$$LHS \text{ of (4)} \lesssim_n S^\varepsilon \prod_{j=1}^n N_j^{\frac{1}{n-1}}.$$

For the general cases, suppose $\rho^{-(M-1)} \leq S \leq \rho^{-M}$. Let Q be a cube containing Q_S of side length ρ^{-M} . Then we have

$$LHS \text{ of (4)} \leq \int_Q \prod_{j=1}^n f_j^{\frac{1}{n-1}}.$$

By the estimation of the case where cubes are of side length ρ^{-M} , and picking $\rho = C_n^{-1/\varepsilon}$, we obtain

$$\int_Q \prod_{j=1}^n f_j^{\frac{1}{n-1}} \lesssim_n \rho^{-\varepsilon} S^\varepsilon \prod_{j=1}^n N_j^{\frac{1}{n-1}} \sim_n S^\varepsilon \prod_{j=1}^n N_j^{\frac{1}{n-1}}.$$

This completes the proof of Lemma 1.2. □

Remark 1.1. The constant $C'_{n,\varepsilon}$ in Lemma 1.2 can be chosen to be independent of ε , but this is not interesting since ultimately we need to cover a cap S_j with many (depending on ε) small caps of small (depending on ε) radius in the proof of Theorem 1.1, forcing the constant $C_{n,\varepsilon}$ in (1) to depend on ε .

2 Multilinear Restriction

The restriction conjecture, also known as the Fourier restriction conjecture, is a conjecture about the behavior of the Fourier transform on curved hypersurfaces. Put $\mathbb{P} : \mathbb{R}^{n-1} \rightarrow \mathbb{R}, x \mapsto |x|^2$, whose graph P is a paraboloid that lives in $\mathbb{R}^n$. We define the **extension operator associated with $\mathbb{P}$** as follows. Given a function

$$g : \mathbb{R}^{n-1} \rightarrow \mathbb{C},$$

put

$$Eg(\xi) = \int_{\mathbb{R}^{n-1}} g(x) e^{i\xi \cdot (x, \mathbb{P}(x))} dx.$$

Eg is a $\mathbb{C}$ -valued function on $\mathbb{R}^n$. The restriction conjecture concerns the boundedness of the operator E from L^p space to L^q space for suitable pairs (p, q) .

Conjecture 2.1. Let $U \subseteq \mathbb{R}^{n-1}$ be a compact neighborhood of 0, $p' \leq \frac{n-1}{n+1}q$ where $q > \frac{2n}{n-1}$. Then we have

$$\|Eg\|_{L^q(\mathbb{R}^n)} \lesssim_{n,p,q} \|g\|_{L^{p'}(U)}, \quad (7)$$

for all $g : U \rightarrow \mathbb{C}$ in $L^{p'}(U)$.

The extension operator can be interpreted as the Fourier inversion of a function on the paraboloid P . Given $g : \mathbb{R}^{n-1} \rightarrow \mathbb{C}$, lift g to

$$\begin{aligned} h : P &\rightarrow \mathbb{C} \\ (x, \mathbb{P}(x)) &\mapsto g(x). \end{aligned}$$

Let σ be the Haar measure pulling back the Lebesgue measure on $\mathbb{R}^{n-1}$ via the natural projection $(x, \mathbb{P}(x)) \mapsto x$. Then we have

$$Eg(\xi) = \int_P h(x) e^{i\xi \cdot x} d\sigma(x) = (h d\sigma)^\vee.$$

One may interpret the study of the extension operator as originating from the intention to study Fourier transforms (on various geometric objects). This is one of the motivations of extension operators. Another comes from the Schrödinger equation:

Given $u(y, t) : \mathbb{R}^{n-1} \times \mathbb{R} \rightarrow \mathbb{C}$, consider the partial differential equation

$$\begin{cases} i\partial_t u - \Delta_y u = 0 \\ u(y, 0) = u_0(y) \end{cases}, \quad (8)$$

where $u_0 : \mathbb{R}^{n-1} \rightarrow \mathbb{C}$. Taking Fourier transform, (7) becomes

$$(\xi_n - |\xi|^2) \hat{u}(\xi, \xi_n) = 0.$$

Hence, if u is a solution of (7), then $\text{supp}(\hat{u}) \subseteq P$. Moreover, $E(\hat{u}_0)$ is a solution of the **Schrödinger equation**.

To deal with the restriction conjecture, the multilinear version is formulated. Let $U_1, \dots, U_n$ be (closed) balls in $B(0, 1) \subseteq \mathbb{R}^{n-1}$, $v > 0$ (think of v as $\frac{1}{10n}$).

Suppose that for all $x_i \in U_i, i = 1, \dots, n$, we have

$$|\det(v(x_1), \dots, v(x_n))| \geq v \text{ (or } |v(x_1) \wedge \dots \wedge v(x_n)| \geq v),$$

where $v(x_1, \dots, x_{n-1}) := (2x_1, \dots, 2x_{n-1}, 1)$. We call such $U_1, \dots, U_n$ v -transverse. Define

$$E_j g_j(\xi) = \int_{U_j} g_j(x) e^{i\xi \cdot (x, \mathbb{P}(x))} dx,$$

for $g_j : U_j \rightarrow \mathbb{C}, j = 1, \dots, n$. The the multilinear restriction conjecture states the following.

Conjecture 2.2. If $p' \leq \frac{n-1}{n}q$, where $q \geq \frac{2n}{n-1}$, then

$$\left\| \prod_{j=1}^n E_j g_j \right\|_{L^{q/n}(\mathbb{R}^n)} \lesssim_{n,p,q,U_1,\dots,U_n,v} \prod_{j=1}^n \|g_j\|_{L^p(U_j)}, \quad (9)$$

for all $g_j : U_j \rightarrow \mathbb{C}, j = 1, \dots, n$.

The transversality condition on $U_1, \dots, U_n$ trivially fails if any two $U_i, U_j, i \neq j$ are too close to each other (or even overlap). When $n \geq 3$, there are also nontrivial cases where distinct U_i, U_j, U_k stand in a line and the transversality fails.

To simplify the problem, pick the endpoint case $q = \frac{2n}{n-1}, p' = \frac{n-1}{n}q = 2$. Then (9) becomes

$$\left\| \prod_{j=1}^n E_j g_j \right\|_{L^{\frac{2}{n-1}}(\mathbb{R}^n)} \lesssim_{n,U_1,\dots,U_n,v} \prod_{j=1}^n \|g_j\|_{L^2(U_j)}, \quad (10)$$

but this is still hard to prove. Simplified further, an essentially easier problem is when the (multilinear) extension operators are orthogonal, in the sense stated below. (This is not a special case of the formulation of Conjecture 2.2, strictly speaking, but essentially can be interpreted as $U_1, \dots, U_n$ being well-spaced to make $v(x_1), \dots, v(x_n)$ almost orthogonal.)

Let $U'_1, \dots, U'_n \subseteq B(0, 1) \subseteq \mathbb{R}^{n-1}$, with no transversality required. For $g_j : U'_j \rightarrow \mathbb{C}$, Define

$$H_j g_j(\xi) = \int_{U_j} g_j(x) e^{i\Pi_j(\xi) \cdot x} dx,$$

where Π_j is the projection from $\mathbb{R}^n$ to $\mathbb{R}^{n-1}$ that forgets the j -th coordinate. Similar to the E_j , H_j can be rewritten as a Fourier inversion:

$$H_j g_j = \check{g}_j \circ \Pi_j.$$

Replacing E_j by H_j in (10), it is natural to expect Loomis-Whitney to find its application in the proof of

Theorem 2.1. Suppose $g_j \in L^2(U'_j)$, $j = 1, \dots, n$. Then the following multilinear-restriction inequality holds:

$$\left\| \prod_{j=1}^n H_j g_j \right\|_{L^{\frac{2}{n-1}}(\mathbb{R}^n)} \leq \prod_{j=1}^n \|g_j\|_{L^2(U'_j)}.$$

Proof. Write

$$\left\| \prod_{j=1}^n H_j g_j \right\|_{L^{\frac{2}{n-1}}(\mathbb{R}^n)}^{\frac{2}{n-1}} = \int_{\mathbb{R}^n} \prod_{j=1}^n |\check{g}_j^{\frac{2}{n-1}} \circ \Pi_j|$$

Note that $\check{g}_j^{\frac{2}{n-1}} \in L^{n-1}$ since $g_j \in L^2$. We can apply Loomis-Whitney to get

$$\int_{\mathbb{R}^n} \prod_{j=1}^n |\check{g}_j^{\frac{2}{n-1}} \circ \Pi_j| \leq \prod_{j=1}^n \|\check{g}_j^{\frac{2}{n-1}}\|_{L^{n-1}} = \prod_{j=1}^n \|g_j\|_{L^2},$$

where the last equality follows from the Plancherel's formula. $\square$

There are two ingredients in the proof of this easy case (the H_j -formulation). One is Plancherel, the other is Loomis-Whitney. Hence, it is not surprising that we shall use multilinear Kakeya in place of Loomis-Whitney in the general cases.

So far we have stated the multilinear conjectures, and have already proved a very special case of the boundedness of multilinear restriction. The following theorem of Bennett-Carbery-Tao is the main result of this section.

Theorem 2.2. For any $\varepsilon > 0$, any (large) positive real number $R (\geq 1)$, we have

$$\left\| \prod_{j=1}^n E_j g_j \right\|_{L^{\frac{2}{n-1}}(B(0,R))} \lesssim_{n,U_1,\dots,U_n,\mathbf{v},\varepsilon} R^\varepsilon \prod_{j=1}^n \|g_j\|_{L^2(U_j)}. \quad (11)$$

Before we embark on the proof of Theorem 2.2, we need a fundamental observation of the uncertainty principle of the Fourier transform.

Lemma 2.1. Suppose that $f \in L^1 \cap L^2(\mathbb{R}^n)$ satisfies $\text{supp } \hat{f} \subseteq B(0,1)$. Then

$$|f(z)| \leq \int_{\mathbb{R}^n} |f(x)| \phi_{B(z,1)}(x) dx, \quad (12)$$

for $z \in \mathbb{R}^n$. Here $\phi_{B(z,1)}(x) := \phi_{B(0,1)}(z-x)$, where $\hat{\phi}_{B(0,1)} = 1$ on $B(0,1)$ is a positive function.

Proof. Note that $\hat{f} = \hat{f} \cdot \hat{\phi}_{B(0,1)}$, we have $f = f * \phi_{B(0,1)}$. Therefore, for any $z \in \mathbb{R}^n$,

$$\begin{aligned} |f(z)| &\leq \int_{\mathbb{R}^n} |f(x)| \phi_{B(0,1)}(z-x) dx \\ &= \int_{\mathbb{R}^n} |f(z)| \phi_{B(z,1)}(x) dx. \end{aligned}$$

$\square$

Remark 2.1. $\phi_{B(0,1)}$ can always be chosen to be a Schwartz function satisfying $\phi_{B(0,1)} \sim 1$ on $B(0,1)$, so that RHS of (12) approximates $\int_{B(z,1)} |f|$. This shows that f is roughly constant on any unit balls whenever its Fourier transform is supported on $B(0,1)$. One can show that f is roughly constant on balls of radius $R \geq 1$ if its Fourier transform is supported on some ball of radius $\frac{1}{R}$, in the same argument as in the proof of Lemma 2.1.

Now we return to the proof of Multilinear Restriction. The idea of our first step is to look at the case when ε in Theorem 2.2 is replaced by fixed positive real numbers α , and then do induction on α . For any $\alpha > 0$, let $\mathcal{R}(\alpha)$ denote the statement that for any $U_1, \dots, U_n \subseteq \mathbb{R}^{n-1}$ that are ν -transverse, $g_j : U_j \rightarrow \mathbb{C}, j = 1, \dots, n, R \geq 1$, we have

$$\left\| \prod_{j=1}^n E_j g_j \right\|_{L^{\frac{2}{n-1}}(B(0,R))} \lesssim C_\alpha R^\alpha \prod_{j=1}^n \|g_j\|_{L^2(U_j)}, \quad (13)$$

for some constant C_α . The implicit constant is of course dependent on $n, U_1, \dots, U_n, \nu$, but from now on we would like to suppress this notation by considering them as fixed. We first prove that $\mathcal{R}(\alpha)$ holds for some large α , and then reach small α by a proper induction.

Proposition 2.1. $\mathcal{R}(\frac{n(n-1)}{2})$ is true.

Proof. We have

$$\begin{aligned} \left\| \prod_{j=1}^n E_j g_j \right\|_{L^{\frac{2}{n-1}}(B(0,R))} &\leq |B(0,R)|^{\frac{n-1}{2}} \left\| \prod_{j=1}^n E_j g_j \right\|_{L^\infty} \\ &\lesssim R^{\frac{n(n-1)}{2}} \prod_{j=1}^n \|g_j\|_{L^1(U_j)} \\ &\lesssim R^{\frac{n(n-1)}{2}} \prod_{j=1}^n \|g_j\|_{L^2(U_j)} \end{aligned}$$

The last inequality follows from Cauchy-Schwartz inequality. $\square$

In the spirit of the uncertainty principle, if a function f changes below the scale of R , then the change of its Fourier transform morally happens above the scale of $1/R$ (Remark 2.1). Therefore, it can be expected that if we blur h_j (the lift of g_j to the paraboloid P) below scale $1/R$ to make them supported on a $\frac{1}{R}$ -neighborhood of a cap on P (so that they really become functions supported on a positive measurable set in $\mathbb{R}^n$), the estimate of the Fourier inversion should be almost the same. Put

$$\mathcal{A}_j = \{(x, |x|^2) : x \in U_j\},$$

and $\mathcal{A}_j^R$ to be the $\frac{1}{R}$ -neighborhood of $\mathcal{A}_j$. For $\alpha > 0$, let $\mathcal{R}'(\alpha)$ denote the statement that for any $U_1, \dots, U_n \subseteq \mathbb{R}^{n-1}$ that are ν -transverse, $f_j : \mathcal{A}_j^R \rightarrow \mathbb{C}, j = 1, \dots, n, R \geq 1$, we have

$$\left\| \prod_{j=1}^n \check{f}_j \right\|_{L^{\frac{2}{n-1}}(B(0,R))} \lesssim C'_\alpha R^{\alpha - \frac{n}{2}} \prod_{j=1}^n \|f_j\|_{L^2(\mathcal{A}_j^R)}, \quad (14)$$

for some constant C'_α .

Lemma 2.2. For $\alpha > 0$, $\mathcal{R}(\alpha)$ is true if and only if $\mathcal{R}'(\alpha)$ is true.

Proof. Suppose $\mathcal{R}'(\alpha)$ is true. To apply $\mathcal{R}'(\alpha)$, pick a Schwartz function ϕ_R such that

$$\begin{cases} \text{supp}(\phi_R) \subseteq B(0, 1/R) \\ \check{\phi}_R|_{B(0,R)} \sim 1 \end{cases}.$$

Then we have $\|\phi_R\|_{L^1} \sim 1$. Given g_j as in the assumption of $\mathcal{R}(\alpha)$, we have

$$\begin{aligned} \left\| \prod_{j=1}^n E_j g_j \right\|_{L^{\frac{2}{n-1}}(B(0,R))} &\sim \left\| \prod_{j=1}^n (\phi_R * h_j d\sigma_j) \right\|_{L^{\frac{2}{n-1}}(B(0,R))} \\ &\stackrel{(\text{by } \mathcal{R}'(\alpha))}{\lesssim} R^{\alpha - \frac{n}{2}} \prod_{j=1}^n \|\phi_R * h_j d\sigma_j\|_{L^2} \\ &\stackrel{(\text{Young's convolution inequality})}{\lesssim} R^{\alpha - \frac{n}{2}} \prod_{j=1}^n \|h_j d\sigma_j\|_{L^2} \lesssim R^\alpha \prod_{j=1}^n \|g_j\|_{L^2(U_j)}, \end{aligned}$$

where h_j defined on $\mathcal{A}_j$ denotes the lift of g_j to the paraboloid. Conversely, suppose $\mathcal{R}(\alpha)$ is true. We can translate $\mathbb{P}$ by $O(1/R)$ and average using Hölder's inequality to finish the proof. Given f_j , then by Hölder's inequality, (RHS of $\mathcal{R}'(\alpha)$)² is

$$\begin{aligned} \prod_{j=1}^n \|f_j\|_{L^2}^2 &= \prod_{j=1}^n \left(\int_{|t| \leq 1/R} dt \int_P |f_j(x + (0, t))|^2 dx \right) \\ &\geq \left(\int_{|t| \leq 1/R} dt \prod_{j=1}^n \left(\int_P |f_j(x + (0, t))| \right)^{1/n} \right)^n. \end{aligned}$$

Put $g_{j,t}(y) = f_j(y, \mathbb{P}(y) + t)$. Then we have $E_j g_{j,t}(\xi', \xi_n) = e^{-i\xi_n t} \check{f}_j(\xi', \xi_n)$, which implies $\|E_j g_{j,t}\|_{L^{\frac{2}{n-1}}(B(0,R))} = \|\check{f}_j\|_{L^{\frac{2}{n-1}}(B(0,R))}$. Therefore, applying $\mathcal{R}(\alpha)$, we obtain

$$\begin{aligned} \left(\int_{|t| \leq 1/R} dt \prod_{j=1}^n \left(\int_P |f_j(x + (0, t))| \right)^{1/n} \right)^n &\gtrsim_\alpha (R^{-\frac{2\alpha}{n}} R^{-1} \left\| \prod_{j=1}^n \check{f}_j \right\|_{L^{\frac{2}{n-1}}(B(0,R))}^{2/n})^n \\ &= R^{-2\alpha - n} \left\| \prod_{j=1}^n \check{f}_j \right\|_{L^{\frac{2}{n-1}}(B(0,R))}^2. \end{aligned}$$

This proves $\mathcal{R}'(\alpha)$. □

Since $\mathcal{R}(\alpha_1)$ implies $\mathcal{R}(\alpha_2)$ if $\alpha_1 \leq \alpha_2$, to prove Theorem 2.2, it suffices to show

Lemma 2.3. For $\alpha > 0$, $\mathcal{R}(\alpha)$ implies $\mathcal{R}(\frac{3}{4}\alpha)$.

Proof. We present the proof in a way that is better motivated.

By Lemma 2, our goal turns to prove $\mathcal{R}'(\frac{3}{4}\alpha)$. Pick a positive Schwartz function ϕ such that

$$\begin{cases} \text{supp} \phi \subseteq B(0, 1), \\ \check{\phi}|_{B(0,1)} \sim 1. \end{cases}$$

For $R \geq 1, x \in \mathbb{R}^n$, define

$$\phi_R^x(\xi) = e^{ix \cdot \xi} R^n \phi(R\xi).$$

Then we have

$$\begin{cases} \text{supp}(\phi_R^x) \subseteq B(0, 1/R), \\ \check{\phi}_R^x|_{B(x, R)} \sim 1. \end{cases}$$

To prove $\mathcal{R}'(\frac{3}{4}\alpha)$, for any $f_j : \mathcal{A}_j^R \rightarrow \mathbb{C}, j = 1, \dots, n$, apply $\mathcal{R}'(\alpha)$ to functions $f_j * \phi_{R^{1/2}}^x$ (supported on $\mathcal{A}_j^{R^{1/2}}$), $j = 1, \dots, n$. We have

$$\left\| \prod_{j=1}^n \check{f}_j \cdot \check{\phi}_{R^{1/2}}^x \right\|_{L^{\frac{2}{n-1}}(B(x, R^{1/2}))} \lesssim C'_\alpha R^{\frac{\alpha}{2} - \frac{n}{4}} \prod_{j=1}^n \|f_j * \phi_{R^{1/2}}^x\|_{L^2(\mathcal{A}_j^{R^{1/2}})}. \quad (15)$$

Average (LHS of (15)) $^{\frac{2}{n-1}}$ over $x \in B(0, R)$,

$$\begin{aligned} \int_{B(0, R)} \left\| \prod_{j=1}^n \check{f}_j \cdot \check{\phi}_{R^{1/2}}^x \right\|_{L^{\frac{2}{n-1}}(B(x, R^{1/2}))}^{\frac{2}{n-1}} dx &= \int_{B(0, R)} \int_{B(x, R^{1/2})} |\phi_{R^{1/2}}^x(y)|^{\frac{2n}{n-1}} \prod_{j=1}^n |\check{f}_j(y)|^{\frac{2}{n-1}} dy dx \\ &\sim \int_{B(0, R)} \int_{B(x, R^{1/2})} \chi_{B(x, R^{1/2})}(y) \prod_{j=1}^n |\check{f}_j(y)|^{\frac{2}{n-1}} dy dx \\ &= \int_{\mathbb{R}^n} \prod_{j=1}^n |\check{f}_j(y)|^{\frac{2}{n-1}} dy \int_{B(0, R)} \chi_{B(y, R^{1/2})}(x) dx \\ &\gtrsim R^{\frac{n}{2}} \int_{B(0, R)} \prod_{j=1}^n |\check{f}_j(y)|^{\frac{2}{n-1}} dy = R^{n/2} \left\| \prod_{j=1}^n \check{f}_j \right\|_{L^{\frac{2}{n-1}}(B(0, R))}^{\frac{2}{n-1}}. \end{aligned}$$

Therefore,

$$\left\| \prod_{j=1}^n \check{f}_j \right\|_{L^{\frac{2}{n-1}}(B(0, R))}^{\frac{2}{n-1}} \lesssim R^{-\frac{n}{2}} R^{(\frac{\alpha}{2} - \frac{n}{4}) \frac{2}{n-1}} \int_{B(0, R)} \prod_{j=1}^n \|f_j * \phi_{R^{1/2}}^x\|_{L^2}^{\frac{2}{n-1}} dx. \quad (16)$$

We analyze f_j locally (U_j is too big compared to $O(R^{-1/2})$), and approach to use multilinear Kakeya as follows. For each j , cover $\mathcal{A}_j^R$ by finitely many bounded (independent of R) overlapping disks $\{\rho_j\}_{\rho_j}$ of radius $R^{-1/2}$ (ρ_j denotes the intersection of the disk with $\text{supp}(f_j)$). Put

$$f_{j, \rho_j} = \chi_{\rho_j} f_j.$$

Then, for each j , we have

$$\sum_{\rho_j} |f_{j, \rho_j}|^2 \lesssim |f_j|^2 \leq \sum_{\rho_j} |f_{j, \rho_j}|^2,$$

which implies

$$\|f_j\|_{L^2}^2 \sim \sum_{\rho_j} \|f_{j, \rho_j}\|_{L^2}^2. \quad (17)$$

Moreover, by the bounded overlap of $\{supp(f_{j,\rho_j} * \phi_{R^{1/2}}^x)\}_{\rho_j}$, we have

$$\|f_j * \phi_{R^{1/2}}^x\|_{L^2}^2 \lesssim \sum_{\rho_j} \|f_{j,\rho_j} * \phi_{R^{1/2}}^x\|_{L^2}^2.$$

Note that $supp(\phi_{R^{1/2}}^x)$ or $supp(f_{j,\rho_j} * \phi_{R^{1/2}}^x)$ is nothing like a tube. To apply multilinear Kakeya, we use Plancherel's formula, so that $\phi_{R^{1/2}}^x$, which is ~ 1 on $B(x, R^{1/2})$ and rapidly decaying off $B(x, R^{1/2})$, can be ignored. In this way, what really needs attention is f_{j,ρ_j} , whose support is a tube of size $O(R^{-1}) \times O(R^{-1/2}) \times \dots \times O(R^{-1/2})$. Therefore, from (16) we obtain

$$\left\| \prod_{j=1}^n \check{f}_j \right\|_{L^{\frac{2}{n-1}}(B(0,R))}^{\frac{2}{n-1}} \lesssim R^{-\frac{n}{2}} R^{\left(\frac{\alpha}{2} - \frac{n}{4}\right) \frac{2}{n-1}} \int_{B(0,R)} \left(\prod_{j=1}^n \sum_{\rho_j} \|f_{j,\rho_j}^{\check{}}\|_{L^2(B(x,R^{1/2}))}^2 \right)^{\frac{1}{n-1}} dx. \quad (18)$$

What remains to prove is easy. Let ρ_j^* denotes the dual set of ρ_j , centered at 0. Write $f_{j,\rho_j} = f'_{j,\rho_j} \cdot \psi_{\rho_j}$, so that $f_{j,\rho_j}^{\check{}} = f'_{j,\rho_j}^{\check{}} * \check{\psi}_{\rho_j}$, where, for each j , ψ_{ρ_j} is a Schwartz function defined to satisfy

$$\begin{cases} \psi_{\rho_j}|_{\rho_j} \sim 1 \\ \check{\psi}_{\rho_j}|_{\rho_j^*} \lesssim R^{-\frac{n+1}{2}}, supp(\check{\psi}_{\rho_j}) \subseteq \rho_j^*. \end{cases}$$

We point out that the constant $R^{-\frac{n+1}{2}}$ is determined by $\psi_{\rho_j}|_{\rho_j} \sim 1$ and Plancherel's formula. Then we obtain, for any $x \in \mathbb{R}^n$,

$$\|f_{j,\rho_j}^{\check{}}\|_{L^2(B(x,R^{1/2}))}^2 \lesssim R^{-\frac{n+2}{2}} (|f'_{j,\rho_j}|^2 * \chi_{\rho_j^*})(x) \quad (19)$$

since we have, $\forall y \in B(0, R^{1/2})$,

$$\begin{aligned} |f_{j,\rho_j}^{\check{}}(x+y)|^2 &\lesssim \int_{\mathbb{R}^n} |f'_{j,\rho_j}(w)|^2 |\check{\psi}_{\rho_j}(x+y-w)|^2 dw \\ &\lesssim R^{-n-1} (|f'_{j,\rho_j}|^2 * \chi_{\rho_j^*})(x) \end{aligned}$$

by expanding the convolution and bounding $\check{\psi}_{\rho_j}$ by $\chi_{\rho_j^*}$. Integrating y over $B(0, R^{1/2})$ gives (19). Note that we may assume $f_j \in L^2$ (otherwise $\mathcal{R}'(\frac{3}{4}\alpha)$ simultaneously holds), so that $|f'_{j,\rho_j}(x)|^2 dx$ is a finite measure on $\mathbb{R}^n$. Therefore we can apply multilinear Kakeya, Plancherel's formula, pointwise comparability of f_{j,ρ_j} , f'_{j,ρ_j} and (17) to have

$$\begin{aligned} \int_{B(0,R)} \left(\prod_{j=1}^n \sum_{\rho_j} (|f_{j,\rho_j}^{\check{}}|^2 * \chi_{\rho_j^*})(x) \right)^{\frac{1}{n-1}} dx &\lesssim_\epsilon R^\epsilon R^{\frac{n(n+1)}{2(n-1)}} \prod_{j=1}^n \left(\sum_{\rho_j} \|f'_{j,\rho_j}\|_{L^2}^2 \right)^{\frac{1}{n-1}} \\ &= R^\epsilon R^{\frac{n(n+1)}{2(n-1)}} \prod_{j=1}^n \left(\sum_{\rho_j} \|f_{j,\rho_j}\|_{L^2}^2 \right)^{\frac{1}{n-1}} \\ &\sim R^\epsilon R^{\frac{n(n+1)}{2(n-1)}} \prod_{j=1}^n \left(\sum_{\rho_j} \|f_j\|_{L^2}^2 \right)^{\frac{1}{n-1}} \\ &\sim R^\epsilon R^{\frac{n(n+1)}{2(n-1)}} \prod_{j=1}^n \|f_j\|_{L^2}^{\frac{2}{n-1}}, \end{aligned}$$

for any $\varepsilon > 0$. Combined with (18), we finally achieve

$$\left\| \prod_{j=1}^n \check{f}_j \right\|_{L^{\frac{2}{n-1}}(B(0,R))}^{\frac{2}{n-1}} \lesssim_{\varepsilon} R^{\varepsilon - \frac{n}{2} + \frac{2}{n-1}(\frac{\alpha}{2} - \frac{n}{4})} \prod_{j=1}^n \|f_j\|_{L^2}^{\frac{2}{n-1}}.$$

Picking $\varepsilon = \frac{n}{2} + \frac{\alpha-n}{2(n-1)} (> 0)$ proves $\mathcal{R}'(\frac{3}{4}\alpha)$. □

One may replace the boundedly overlapping $\{\rho_j\}_{\rho_j}$ by disjoint rectangle-like sets of size $O(R^{-1}) \times O(R^{-1/2}) \times \cdots \times O(R^{-1/2})$ in the proof of Lemma 2.3, and the argument still works.

3 Linear Kakeya

Suppose as usual $n \geq 2$, $\delta \in (0,1)$. Let k be a positive integer with $1 \leq k \leq n$. Suppose $\mathbb{T}_1, \dots, \mathbb{T}_k$ are (finite) sets of δ -tubes in $\mathbb{R}^n$, with tubes in $\mathbb{T}_j$ pointing in directions making angle $\leq \frac{1}{100}n$ with e_j . Then, for any $\varepsilon > 0$ we have the k -linear Kakeya estimate

$$\left\| \prod_{j=1}^k \left(\sum_{T_j \in \mathbb{T}_j} \chi_{T_j} \right)^{1/k} \right\|_{L^{\frac{k}{k-1}}} \lesssim_{n,\varepsilon} \delta^{1-\frac{n}{k}-\varepsilon} \prod_{j=1}^k (\delta^{n-1} \# \mathbb{T}_j)^{1/k}. \quad (20)$$

The case $k = n$ recovers the exact statement of Theorem 1.1 (BCT). The case $k = 1$ is trivial by the fact $\|\sum_{T_1 \in \mathbb{T}_1} \chi_{T_1}\| \leq \# \mathbb{T}_1$. In this section, we will use multilinear Kakeya, together with the technique of induction on scales, to prove the following linear Kakeya theorem.

Theorem 3.1. *Let $1 \leq d \leq n$, $\mathbb{T}$ a set of δ -tubes in $\mathbb{R}^n$ pointing in δ -separated directions (which implies $\# \mathbb{T} \lesssim 1/\delta^{n-1}$). If*

$$d = \begin{cases} \frac{n}{2} + 1, & n \text{ even} \\ \frac{n+1}{2}, & n \text{ odd} \end{cases},$$

then we have, for any $\varepsilon > 0$,

$$\left\| \sum_{T \in \mathbb{T}} \chi_T \right\|_{L^{\frac{d}{d-1}}(\mathbb{R}^n)} \lesssim_{n,\varepsilon} \delta^{-\varepsilon} \delta^{1-\frac{n}{d}} (\delta^{n-1} \# \mathbb{T})^{\frac{d-1}{d}}. \quad (21)$$

Regarding Kakeya Conjecture, we have an immediate corollary.

Corollary 3.1. *Let $1 \leq d \leq n$. If (21) holds for d , then every Besicovitch set E in $\mathbb{R}^n$ has (lower) Minkowski dimension $\dim_M(E) \geq d$. In particular, Theorem 3.1 resolves the 2-dimensional Kakeya.*

Proof. For any $\delta > 0$, denote by E^δ the δ -neighborhood of E . Then E^δ includes a δ -tube in every δ -separated direction. Collect them as $\mathbb{T}$. By Hölder's inequality, we have

$$1 \sim \left\| \sum_{T \in \mathbb{T}} \chi_T \right\|_{L^1} \leq \left\| \sum_{T \in \mathbb{T}} \chi_T \right\|_{L^{\frac{d}{d-1}}} \|\chi_{E^\delta}\|_{L^d},$$

and thus $|E^\delta| \lesssim_\varepsilon \delta^{-\varepsilon} \delta^{1-\frac{n}{d}}$, for any $\varepsilon > 0$. Therefore we have

$$\liminf_{\delta \rightarrow 0} \frac{\log N(\delta)}{\log(1/\delta)} \geq d - \varepsilon,$$

where $N(\delta) \sim |E^\delta|/\delta^n$ is the covering number of E^δ by δ -boxes. Letting ε tend to 0 completes the proof. $\square$

Therefore, if (21) is justified for $d = n$, then the (Minkowski-version) Kakeya conjecture is resolved. In 1995, Wolff [3] proved that every Besicovitch set in $\mathbb{R}^n$ has Hausdorff

dimension no less than $\frac{n+2}{2}$ by giving an estimate of the Keakeya-type maximal operator, which naturally reveals the geometry of intersection of tubes.

To prove Theorem 3.1, a few notations concerning broad-narrow analysis are needed for the convenience of later arguments. From now on till the end of this section we argue under the assumptions of Theorem 3.1, unless stated otherwise.

Let $K > 0$ be a large (undetermined) real number, allowed to depend on n, ε but not on δ . (Imagine $K = 1000^{\frac{1}{\varepsilon}}$.) Cover S^{n-1} with boundedly overlapping caps $\{\tau\}_\tau$ of diameter $1/K$. For each cap τ , write

$$\mathbb{T}_\tau := \{T \in \mathbb{T} : \text{dir}(T) \in \tau\}.$$

Then by definition, we have

$$\#\mathbb{T} \sim \sum_{\tau} \#\mathbb{T}_\tau, \quad \#\{\tau\}_\tau \sim K^{n-1}.$$

Definition 3.1. Suppose $x \in \bigcup_{T \in \mathbb{T}} T$. We say a cap τ is significant for x if

$$\sum_{T \in \mathbb{T}_\tau} \chi_T(x) \geq K^{-100n} \sum_{T \in \mathbb{T}} \chi_T(x),$$

and let $S(x)$ be the set of all significant caps for x , which must be nonempty.

By definition of significant caps, for any $x \in \bigcup_{T \in \mathbb{T}} T$, we must have

$$\sum_{T \in \mathbb{T}} \chi_T(x) \sim \sum_{\tau \in S(x)} \sum_{T \in \mathbb{T}_\tau} \chi_T(x).$$

Definition 3.2. We say a point $x \in \bigcup_{T \in \mathbb{T}} T$ is k -broad if there exist k significant caps for x that are not included in the $100/K$ -neighborhood of any $(k-1)$ -dimensional subspace of $\mathbb{R}^n$. Otherwise we call x a k -narrow point. Denote by X_{broad} the set of k -broad points, by X_{narrow} the set of k -narrow points.

Then we can rewrite LHS of (21) by

$$\left\| \sum_{T \in \mathbb{T}} \chi_T \right\|_{L^{\frac{d}{d-1}}}^{\frac{d}{d-1}} = \left\| \sum_{T \in \mathbb{T}} \chi_T \right\|_{L^{\frac{d}{d-1}}(X_{\text{broad}})}^{\frac{d}{d-1}} + \left\| \sum_{T \in \mathbb{T}} \chi_T \right\|_{L^{\frac{d}{d-1}}(X_{\text{narrow}})}^{\frac{d}{d-1}}.$$

Take $k = d$. For the broad part, we have

Proposition 3.1. For $1 \leq d \leq n$, the following inequality holds.

$$\left\| \sum_{T \in \mathbb{T}} \chi_T \right\|_{L^{\frac{d}{d-1}}(X_{\text{broad}})}^{\frac{d}{d-1}} \lesssim_{n, \varepsilon} K^{O(n^2)} \delta^{\frac{d-n-\varepsilon}{d-1}} (\delta^{n-1} \#\mathbb{T}) \quad (22)$$

Proof. If $x \in X_{\text{broad}}$, caps $\tau_1, \dots, \tau_k$ significant for x and not included in the neighborhood of any $(k-1)$ -dimensional subspace of $\mathbb{R}^n$, we have

$$\prod_{j=1}^k \left(\sum_{T_j \in \mathbb{T}_{\tau_j}} \chi_{T_j}(x) \right)^{1/k} \geq K^{-100n} \sum_{T \in \mathbb{T}} \chi_T(x). \quad (23)$$

For various $\tau_1, \dots, \tau_k$ transversal as such, consider

$$X_{broad}^{\tau_1, \dots, \tau_k} := \{x \in X_{broad} : \tau_1, \dots, \tau_k \in S(x)\}.$$

By pigeonholing, there exist caps $\tau_1, \dots, \tau_k$ not included in any $100/K$ -neighborhood of any $(k-1)$ -dimensional subspace such that

$$\left\| \sum_{T \in \mathbb{T}} \chi_T \right\|_{L^{\frac{d}{d-1}}(X_{broad})}^{\frac{d}{d-1}} \lesssim_n K^{k(n-1)} \left\| \sum_{T \in \mathbb{T}} \chi_T \right\|_{L^{\frac{d}{d-1}}(X_{broad}^{\tau_1, \dots, \tau_k})}^{\frac{d}{d-1}}. \quad (24)$$

Here we point out that the constant $K^{k(n-1)}$ comes from the number of choices of k caps from $\{\tau\}_\tau$. Suppress the notation $X_{broad}^{\tau_1, \dots, \tau_k}$ as X'_{broad} . Taking $k = d$, put (24)(23) and d -linear Kakeya together. We have

$$\begin{aligned} \left\| \sum_{T \in \mathbb{T}} \chi_T \right\|_{L^{\frac{d}{d-1}}(X_{broad})}^{\frac{d}{d-1}} &\lesssim_n K^{d(n-1)} \left\| \sum_{T \in \mathbb{T}} \chi_T \right\|_{L^{\frac{d}{d-1}}(X'_{broad})}^{\frac{d}{d-1}} \\ &\leq K^{d(n-1)} K^{100 \frac{d}{d-1} n} \left\| \prod_{j=1}^d \sum_{T_j \in \mathbb{T}_{\tau_j}} \chi_{T_j} \right\|_{L^{\frac{d}{d-1}}(X'_{broad})}^{\frac{d}{d-1}} \\ &\lesssim_{n, \varepsilon} K^{O(n^2)} \delta^{\frac{d-n-\varepsilon}{d-1}} \prod_{j=1}^d (\delta^{n-1} \# \mathbb{T}_{\tau_j})^{\frac{1}{d-1}} \\ &\leq K^{O(n^2)} \delta^{\frac{d-n-\varepsilon}{d-1}} (\delta^{n-1} \# \mathbb{T})^{\frac{d}{d-1}} \lesssim_n K^{O(n^2)} \delta^{\frac{d-n-\varepsilon}{d-1}} (\delta^{n-1} \# \mathbb{T}), \end{aligned}$$

completing the proof of the broad part, where we are indeed allowed to apply d -linear Kakeya to $\{\mathbb{T}_j\}_{j=1}^d$ due to an invertible linear transformation converting every $\mathbb{T}_j$ to a family of tubes pointing in the directions $100/K$ -close to e_j . $\square$

For the narrow part, note that for $x \in X_{narrow}$, all significant caps are included in the $1000/K$ -neighborhood of a $(k-1)$ -dimensional subspace of $\mathbb{R}^n$, and thus included in a $(k-2)$ -dimensional great circle of S^{n-1} . (We may assume $k (= d) \geq 2$, since the case $d = 1, n \geq 2$ of (21) trivially holds.) Therefore we have $\#S(x) \lesssim K^{k-2}$. Therefore, for $x \in X_{narrow}$, by Hölder's inequality we have

$$\begin{aligned} \left(\sum_{T \in \mathbb{T}} \chi_T(x) \right)^{\frac{d}{d-1}} &\sim \left(\sum_{\tau \in S(x)} \sum_{T \in \mathbb{T}_\tau} \chi_T(x) \right)^{\frac{d}{d-1}} \\ &\leq (\#S(x))^{\frac{1}{d-1}} \sum_{\tau} \left(\sum_{T \in \mathbb{T}_\tau} \chi_T(x) \right)^{\frac{d}{d-1}} \\ &\lesssim_n K^{\frac{d-2}{d-1}} \sum_{\tau} \left(\sum_{T \in \mathbb{T}_\tau} \chi_T(x) \right)^{\frac{d}{d-1}}. \end{aligned}$$

For tubes pointing into the same cap, those which intersect on almost nothing, we can, and indeed should, treat them separately. For any cap τ , let $\hat{\mathbb{T}}_\tau = \{\hat{T}_\tau\}$ be a family of boundedly overlapping $O(1/K)$ -tubes pointing exactly at the center of τ , such that every $T_\tau \in \mathbb{T}_\tau$ is contained in some $\hat{T}_\tau \in \hat{\mathbb{T}}_\tau$. Put $\hat{\mathbb{T}} = \bigcup_\tau \hat{\mathbb{T}}_\tau$. For any $\hat{T} \in \hat{\mathbb{T}}$, define

$$\mathbb{T}[\hat{T}] = \{T \in \mathbb{T}_\tau : T \subseteq \hat{T}\}$$

if $\hat{T}$ points at the center of τ . Then for every τ , any $x \in X_{\text{narow}}$, we have

$$\begin{aligned} \left(\sum_{T \in \mathbb{T}_\tau} \chi_T(x) \right)^{\frac{d}{d-1}} &\leq \left(\sum_{\hat{T}_\tau \in \hat{\mathbb{T}}_\tau} \sum_{T \in \mathbb{T}[\hat{T}_\tau]} \chi_T(x) \right)^{\frac{d}{d-1}} \\ (\text{bouded overlap of } \hat{\mathbb{T}}_\tau) &\lesssim \sum_{\hat{T}_\tau \in \hat{\mathbb{T}}_\tau} \left(\sum_{T \in \mathbb{T}[\hat{T}_\tau]} \chi_T(x) \right)^{\frac{d}{d-1}}. \end{aligned}$$

Therefore,

$$\left(\sum_{T \in \mathbb{T}} \chi_T(x) \right)^{\frac{d}{d-1}} \lesssim_n K^{\frac{d-2}{d-1}} \sum_{\tau} \sum_{\hat{T}_\tau \in \hat{\mathbb{T}}_\tau} \left(\sum_{T \in \mathbb{T}[\hat{T}_\tau]} \chi_T(x) \right)^{\frac{d}{d-1}} = K^{\frac{d-2}{d-1}} \sum_{\hat{T} \in \hat{\mathbb{T}}} \left(\sum_{T \in \mathbb{T}[\hat{T}]} \chi_T(x) \right)^{\frac{d}{d-1}}. \quad (25)$$

Integrating (25) over X_{narow} , we obtain

$$\left\| \sum_{T \in \mathbb{T}} \chi_T \right\|_{L^{\frac{d}{d-1}}(X_{\text{narow}})}^{\frac{d}{d-1}} \lesssim_n K^{\frac{d-2}{d-1}} \sum_{\hat{T} \in \hat{\mathbb{T}}} \left\| \sum_{T \in \mathbb{T}[\hat{T}]} \chi_T \right\|_{L^{\frac{d}{d-1}}(X_{\text{narow}})}^{\frac{d}{d-1}}.$$

Note that $\#\hat{\mathbb{T}} \sim K^{2(n-1)}$ and our goal is to get an estimate

$$\left\| \sum_{T \in \mathbb{T}[\hat{T}]} \chi_T \right\|_{L^{\frac{d}{d-1}}(X_{\text{narow}})}^{\frac{d}{d-1}} \lesssim_{n,\varepsilon} \delta^{\frac{d-n-\varepsilon}{d-1}} (\delta^{n-1} \#\mathbb{T}), \quad \forall \hat{T} \in \hat{\mathbb{T}}. \quad (26)$$

For then we would, combined with Proposition 3.1, complete the proof of Theorem 3.1. We prove (26) by induction on δ for the conclusion of Theorem 3.1. The base of our induction

$$\left\| \sum_{T \in \mathbb{T}} \chi_T \right\|_{L^{\frac{d}{d-1}}}^{\frac{d}{d-1}} \lesssim_n \#\mathbb{T}$$

trivially holds, where $\delta \sim 1$ and thus $\#\mathbb{T} \sim 1$. Therefore, to prove (26) we may assume that (21) holds with δK in place of δ . Now we rescale our original δ -tubes. Fix any $\hat{T}$ and put $f = \sum_{T \in \mathbb{T}[\hat{T}]} \chi_T$, $\tilde{f}(x) := f(K^{-1}x)$. We have

$$\begin{aligned} \|f\|_{L^{\frac{d}{d-1}}(X_{\text{narow}})}^{\frac{d}{d-1}} &= K^{-n} \|\tilde{f}\|_{L^{\frac{d}{d-1}}(KX_{\text{narow}})}^{\frac{d}{d-1}} \\ &\lesssim_{n,\varepsilon} K^{-n} (\delta K)^{\frac{d-n-\varepsilon}{d-1}} ((\delta K)^{n-1} \#\mathbb{T}) \\ &\leq C_K \delta^{\frac{d-n-\varepsilon}{d-1}} (\delta^{n-1} \#\mathbb{T}), \end{aligned}$$

where $C_K = K^{-\frac{\varepsilon}{d-1}} K^{\frac{d-n}{d-1}-1}$. This closes the induction and therefore the proof of Theorem 3.1.

4 Restriction Estimates

In this section, we mainly introduce the work of Guth / Bourgain-Guth. Let $n \geq 3$, $2 \leq k \leq n$, $1 \leq p < \infty$, and E denotes the extension operator defined in the previous section.

Fix a positive integer $A \sim 100$. Pick any $K > 10$, and cover $B(0, 1) \subseteq \mathbb{R}^{n-1}$ with finitely overlapping disks $\{\tau\}$ of radius K^{-1} , with the center of each disk τ denoted by w_τ . Let v_τ denote the normal vector of the paraboloid P at $(w_\tau, \mathbb{P}(\tau))$. For $g : B(0, 1) \rightarrow \mathbb{C}$ and any ball $B_{K^2} \subseteq \mathbb{R}^n$ of radius K^2 , put

$$\mu_{Eg}^p(B_{K^2}) = \min_{V_1, \dots, V_A \in Gr(k-1, n)} \max_{\substack{\tau: \angle(v_\tau, V_j) \geq 100K^{-1} \\ \forall j=1, \dots, A}} \|E(g|_\tau)\|_{L^p(B_{K^2})}^p, \quad (27)$$

where $Gr(k-1, n)$ denotes the set of $(k-1)$ -dimensional subspaces of $\mathbb{R}^n$, $\angle(v, V) := \min_{u \in V} \angle(v, u)$ the least angle of v with nonzero vectors in V . Let B_R be a ball of (large) radius R covered by finitely overlapping balls B_{K^2} of radius K^2 . Define the **k -broad norm** to be

$$\|Eg\|_{BL_{k,A}^p(B_R)}^p := \sum_{B_{K^2}} \mu_{Eg}^p(B_{K^2}).$$

Then we can show that estimates of the k -broad norm implies restriction estimates for certain range of p .

Theorem 4.1. *Suppose that for any $K > 10$, any small $\varepsilon > 0$, we have*

$$\|Eg\|_{BL_{k,A}^p(B_R)} \lesssim_{K,p,n,\varepsilon} R^\varepsilon \|g\|_{L^p}. \quad (28)$$

Then, if $\frac{2(n+1)}{n-1} \geq p \geq 2 + \frac{4}{2n-k}$, we have

$$\|Eg\|_{L^p(B_R)} \lesssim_{n,p,\varepsilon} R^\varepsilon \|g\|_{L^p}. \quad (29)$$

In the following argument to the end of this section, we consider K, p, n as fixed and suppress the notation of dependence of constants on K, p, n . (Imagine $K = 50$.)

Proof. Fix $g : B(0, 1) \rightarrow \mathbb{C}$. Fix a covering $\{B_{K^2}\}$ of B_R . For each B_{K^2} , fix the choice of $V_1, \dots, V_A$ that reaches the minimum in (27). Put

$$N(B_{K^2}) = \{\tau : \exists j \in \{1, \dots, A\}, \angle(v_\tau, V_j) < 100K^{-1}\}.$$

Then we have

$$\mu_{Eg}^p(B_{K^2}) = \max_{\tau \notin N(B_{K^2})} \int_{B_{K^2}} |Eg_\tau|^p,$$

where g_τ denotes $g|_\tau$. Using discrete Hölder's inequality, we obtain

$$\begin{aligned} \|Eg\|_{L^p(B_{K^2})}^p &\leq 2^{p-1} (\#\{\tau\})^{p-1} \sum_{\tau \notin N(B_{K^2})} \int_{B_{K^2}} |Eg_\tau|^p + 2^{p-1} \int_{B_{K^2}} \left| \sum_{\tau \in N(B_{K^2})} Eg_\tau \right|^p \\ &\lesssim K^{(n-1)p} \mu_{Eg}^p(B_{K^2}) + \int_{B_{K^2}} \left| \sum_{\tau \in N(B_{K^2})} Eg_\tau \right|^p. \end{aligned}$$

We will be able to deal with μ_{Eg}^p using the assumed estimate of (28). For the narrow part, by Bourgain's l^2 -decoupling inequality, we have

$$\left\| \sum_{\tau \in N(B_{K^2})} Eg\tau \right\|_{L^p(B_{K^2})}^2 \lesssim_\varepsilon K^\varepsilon \sum_{\tau \in N(B_{K^2})} \|Eg\tau\|_{L^p(B_{K^2})}^2, \quad (30)$$

and therefore

$$\begin{aligned} \int_{B_{K^2}} \left| \sum_{\tau \in N(B_{K^2})} Eg\tau \right|^p &= \left\| \sum_{\tau \in N(B_{K^2})} Eg\tau \right\|_{L^p(B_{K^2})}^p \\ &\stackrel{(by (30))}{\lesssim_\varepsilon} (K^\varepsilon \sum_{\tau \in N(B_{K^2})} \|Eg\tau\|_{L^p(B_{K^2})}^2)^{p/2} \\ &\leq K^{\varepsilon p/2} (\#N(B_{K^2}))^{\frac{p}{2}-1} \sum_{\tau \in N(B_{K^2})} \|Eg\tau\|_{L^p(B_{K^2})}^p \\ &\lesssim K^{\varepsilon p/2} A^{\frac{p}{2}-1} K^{(k-2)(\frac{p}{2}-1)} \sum_{\tau \in N(B_{K^2})} \|Eg\tau\|_{L^p(B_{K^2})}^p. \end{aligned}$$

Therefore, summing over balls $\{B_{K^2}\}$, we obtain

$$\|Eg\|_{L^p(B_R)}^p \lesssim_\varepsilon K^{(n-1)p} \|Eg\|_{BL_{k,A}^p}^p + A^{\frac{p}{2}-1} K^{\varepsilon p/2} K^{(k-2)(\frac{p}{2}-1)} \sum_{\tau} \|Eg\tau\|_{L^p(B_R)}^p \quad (31)$$

$$\stackrel{(by (27))}{\lesssim_\varepsilon} K^{(n-1)p} R^{\varepsilon p} \|g\|_{L^p}^p + A^{\frac{p}{2}-1} K^{\varepsilon p/2} K^{(k-2)(\frac{p}{2}-1)} \sum_{\tau} \|Eg\tau\|_{L^p(B_R)}^p. \quad (32)$$

Now Fix any cap τ and look at $\|Eg\tau\|_{L^p(B_R)}^p$. We can prove the whole theorem by induction on R and actually do the induction by rescaling. By translation, we may suppose τ is centered at 0. Knowing that the base of the induction

$$\int_{B_1} |Eg|^p \lesssim \|g\|_{L^p}^p$$

trivially holds, we can assume our induction hypothesis as

$$\int_{B_{R/K}} |Eg|^p \lesssim (R/K)^\varepsilon \|g\|_{L^p}^p \quad (33)$$

for any τ' of any fixed radius (imagine radius 1).

To deduce the R -case, put

$$\tilde{g}_\tau(x) = g_\tau(K^{-1}x).$$

Then we have $|Eg_\tau(\xi)| = K^{1-n} |E\tilde{g}_\tau(\tilde{\xi})|$, $\forall \xi \in B_R$, where $\tilde{\xi} := (K^{-1}\xi', K^{-2}\xi_n)$ is the parabolic rescaling of ξ . Hence, with (33) at hand, we obtain

$$\begin{aligned} \int_{B_R} |Eg_\tau|^p &\leq K^{(1-n)p} K^{n+1} \int_{R/K} |E\tilde{g}_\tau|^p \\ &\lesssim_\varepsilon R^\varepsilon K^{-\varepsilon} K^{(1-n)p+n+1} \|\tilde{g}_\tau\|_{L^p}^p \\ &= R^\varepsilon K^{-\varepsilon} K^{(1-n)p+2n} \|g_\tau\|_{L^p}^p. \end{aligned}$$

Recall (32), we know that the induction closes if

$$(k-2)\left(\frac{p}{2}-1\right)+(1-n)p+2n \leq 0$$

which is equivalent to $p \geq 2 + \frac{4}{2n-k}$. The proof of Theorem 4.1 is now complete. $\square$

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